

A Partial Answer to Baudier's $C(K)$ -Universality Question

agent_lane_06

2026-06-29

Status

partial_result_likely_valid. This packet answers the isometric-universality clause of Question 2 in arXiv:1305.4025 for compact metric spaces, and gives the uncountable compact metric side of the $c_{C(K)} = 1$ question. It does not settle the remaining almost-isometric $c_{C(K)} = 1$ problem for countable compact metric spaces of high Cantor-Bendixson rank.

Source Question

The source is Florent Baudier, *A topological obstruction for small-distortion embeddability into spaces of continuous functions on countable compact metric spaces*, arXiv:1305.4025. In Section 3, Question 2 asks:

Characterize compacta K such that $c_{C(K)}(\text{SEP}) = 1$

or $c_{C(K)}(\mathcal{SEP}) = 1$, and asks for which K the space $C(K)$ is an isometric universal Banach space for the class of separable Banach spaces.

Idea of the Proof

The two sides are separated by the perfect-set theorem. An uncountable compact metric space contains a closed Cantor set. The Cantor set maps continuously onto $[0, 1]$, so $C([0, 1])$ embeds isometrically into C of that Cantor set by pullback. The Borsuk-Dugundji extension theorem then embeds that C -space isometrically into $C(K)$. Thus the Banach-Mazur theorem transfers the usual universality of $C([0, 1])$ to $C(K)$.

For countable compact metric K , the dual $C(K)^*$ is separable, since measures on the scattered countable compactum are atomic and form $\ell_1(K)$. Hence no subspace of $C(K)$ can be linearly isomorphic to ℓ_1 . Godefroy-Kalton's isometric lifting theorem converts a metric isometric copy of ℓ_1 in any Banach space into a linear isometric copy, so countable compacta are excluded.

Theorem 1. *Let K be a compact metric space. Then $C(K)$ is an isometric universal Banach space for the class of separable Banach spaces if and only if K is uncountable.*

In addition, if K is uncountable compact metric, then

$$c_{C(K)}(\text{SEP}) = c_{C(K)}(\mathcal{SEP}) = 1.$$

Proof. Assume first that K is uncountable compact metric. By the classical perfect-set theorem, K contains a closed subset F homeomorphic to the Cantor set $2^{\mathbb{N}}$. There is a continuous surjection $q : F \rightarrow [0, 1]$. Therefore

$$J : C([0, 1]) \rightarrow C(F), \quad Jf = f \circ q,$$

is a linear isometry, because $q(F) = [0, 1]$.

By the Borsuk-Dugundji extension theorem, since F is closed in the compact metric space K , there is a linear extension operator

$$E : C(F) \rightarrow C(K)$$

with $\|E\| = 1$ and $(Eg)|_F = g$ for every $g \in C(F)$. It follows that E is an isometry: $\|Eg\|_{C(K)} \geq \|g\|_{C(F)}$ by restriction to F , while $\|Eg\| \leq \|g\|$ by $\|E\| = 1$. Hence EJ is a linear isometric embedding of $C([0, 1])$ into $C(K)$.

The Banach-Mazur theorem says that every separable Banach space embeds linearly isometrically into $C([0, 1])$, so every separable Banach space embeds linearly isometrically into $C(K)$. Thus $C(K)$ is isometrically universal for separable Banach spaces. Also, every separable metric space embeds isometrically into a separable Lipschitz-free space, and this Banach space embeds linearly isometrically into $C([0, 1])$. Hence every separable metric space embeds isometrically into $C(K)$, proving $c_{C(K)}(\text{SEP}) = c_{C(K)}(\mathcal{SEP}) = 1$.

Conversely, suppose K is countable compact metric. Then K is scattered, and every regular Borel measure on K is purely atomic. Thus $C(K)^*$ is isometric to $\ell_1(K)$, which is separable because K is countable. If $Y \subseteq C(K)$ is a closed subspace, then Y^* is a quotient of $C(K)^*$, so Y^* is separable. Therefore $C(K)$ cannot contain a linear subspace isomorphic to ℓ_1 , since $\ell_1^* = \ell_\infty$ is nonseparable.

If $C(K)$ were isometrically universal for separable Banach spaces, then ℓ_1 would embed isometrically into $C(K)$ as a metric space. By the theorem of Godefroy and Kalton, whenever a separable Banach space embeds isometrically into a Banach space, it is linearly isometric to a subspace of that Banach space. This would give a linear isometric copy of ℓ_1 in $C(K)$, contradicting the previous paragraph. Hence countable compact metric K are not isometric universal Banach compacta. $\square$

Limitations

This theorem does not decide whether there are countable compact metric spaces K with $c_{C(K)}(\text{SEP}) = 1$ or $c_{C(K)}(\mathcal{SEP}) = 1$ in the almost-isometric sense. Baudier's Corollary 2.4 only forces $\sigma(K) \geq \omega$ for such examples; the countable high-rank case remains outside this packet.

Novelty and Search Notes

The lightweight run indexes had no exact entry for arXiv:1305.4025. Local and web searches for the source title, $c_{C(K)}(\text{SEP})$, and $C(K)$ -distortion did not reveal a later exact packet or answer in this run. The argument uses standard classical ingredients: the perfect-set theorem for uncountable compact metric spaces, the Borsuk-Dugundji extension theorem, Banach-Mazur universality of $C([0, 1])$, and the Godefroy-Kalton theorem on isometric embeddings of separable Banach spaces.

References

- F. Baudier, *A topological obstruction for small-distortion embeddability into spaces of continuous functions on countable compact metric spaces*, arXiv:1305.4025.
- S. Banach and S. Mazur, *Zur Theorie der linearen Dimension*, Studia Math. 4 (1933), 100–112.
- J. Dugundji, *An extension of Tietze's theorem*, Pacific J. Math. 1 (1951), 353–367.

- G. Godefroy and N. J. Kalton, *Lipschitz-free Banach spaces*, Studia Math. 159 (2003), 121–141.